

QuickSort trace

```
quickSort(a, 0, 7)
```

```
1. pivotIndex = partition(a, 0, 7)
```

		0	1	2	3	4	5	6	7	
first	0	3	5	0	4	6	1	2	4	4

last	7	1			6	7
		iFL			iFR	

0	1	2	3	4	5	6	7
3	2	0	4	6	1	5	4

3	5
iFL	iFR

0	1	2	3	4	5	6	7
3	2	0	1	6	4	5	4

3	4
iFR	iFL

0	1	2	3	4	5	6	7
3	2	0	1	4	4	5	6

smaller

1
iFL

larger

pivotIndex =

4

```
2. quickSort(a, 0, 3)
```

```
2.1 pivotIndex = partition(a, 0, 3)
```

first	0					pivot	1
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last	3
pivotIndex	3

0	1	2	3	4	5	6	7
3	2	0	1	4	4	5	6

0	2
iFL	iFR

0	1	2	3	4	5	6	7
0	2	3	1	4	4	5	6

0	1
---	---

iFR	iFL						
0	1	2	3	4	5	6	7
0	1	3	2	4	4	5	6

1

iFL

pivotIndex = 1

2.2 quickSort(a, 0, 0)

Base case: first == last

0	1	2	3	4	5	6	7
0	1	3	2	4	4	5	6

2.3 quickSort(a, 2, 3)

2.3.1 pivotIndex = partition(a, 2, 3)

first 2

pivot 2

last 3

pivotIndex 3

0	1	2	3	4	5	6	7
0	1	3	2	4	4	5	6

2 2

iFR iFL

0	1	2	3	4	5	6	7
0	1	2	3	4	4	5	6

2

iFL

pivotIndex = 2

2.3.2 quickSort(a, 2, 1)

Base case: first > last

0	1	2	3	4	5	6	7
0	1	2	3	4	4	5	6

2.3.3 quickSort(a, 3, 3)

Base case: first == last

0	1	2	3	4	5	6	7
0	1	2	3	4	4	5	6

3. quickSort(a, 5, 7

3.1 pivotIndex = par

first 5

last 7

0	1	2	3	4	5	6	7
0	1	2	3	4	4	5	6

6	7
iFR	iFL

0	1	2	3	4	5	6	7
0	1	2	3	4	4	5	6

7
iFL

pivotIndex = 7

3.2 quickSort(a, 5,

3.2.1 pivotIndex = p

[illegible]

[illegible]